

CALRIG · REQUIREMENTS

CalRig requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/calrig		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.9	2026-10-02	Amish Chadha	Statuses from CLR-CAL-001 v0.7: R10 met including the large item case (was met for six heads only); R12 over the value-engineering target by USD 36.50 (was USD 27); R13 figures 383 mm and 13.9 kg, still met; R14 report template defined, still not verifiable at TRL 3
0.8	2026-10-02	Amish Chadha	Decisions of 2026-10-02 (CLR-DEC-001): R8 first candidate collocation site named; R10 large item case added; R14 report based on the US EPA 2021 targets and protocol with an uncertainty chain. No status changed
0.7	2026-10-01	Amish Chadha	Budget treated as a value-engineering target; cost wording only, no number changed
0.6	2026-10-01	Amish Chadha	Statuses from CLR-CAL-001 v0.4 for the constructable design (CLR-DDR-003); R12 Met to Not met (\$439), budget rise proposed; R13 figures updated

0.5	2026-09-26	Amish Chadha	Budget approved by Amish; R12 target \$412, status Not met to Met
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R12 target \$400, R13 mass relaxed to 14 kg, statuses from CLR-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	R1 redefined per CLR-DDR-001 A3; status column replaced by the CLR-CAL-001 results; budget note per A1
0.3	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with first-order status
0.1	2026-09-25	Amish Chadha	Initial scaffold

CalRig requirements

These requirements have been checked by calculation in CLR-CAL-001 v0.7. Targets are not yet validated with users. R1 was redefined under CLR-DDR-001 (item A3); R12 and R13 were restated under CLR-DDR-002. All three changes were decided by Amish on 2026-09-25 (go with recommendation). Test conditions follow the enhanced lab conditions in the US EPA sensor testing reports (20 °C and 40 °C; 40 % and 85 % RH) so that CalRig results can be read against a known method ([EPA FAQ](#)).

The **reference room** is an indoor lab at 22 °C and 50 % RH unless a requirement states otherwise.

Table 1. Requirements with targets and status from CLR-CAL-001 (Table 4 there gives the values and script tags).

ID	Requirement	Target	Verification (TRL 3 or later)	Status (CLR-CAL-001)
R1	Chamber temperature range	10 to 40 °C at the sensor tray in a room at 15 to 25 °C; in rooms from 25 to 30 °C, the lowest reachable set point is stated in each calibration record	Heat balance with a thermoelectric model; later chamber log	Met: 6.7 °C reachable in a 25 °C room; 11.1 °C lowest in a 30 °C room
R2	Chamber humidity range	20 to 85 % RH at any temperature from 20 to 40 °C	Moisture balance; later chamber log	Met: all corners reached in 26 min or less; with the larger inner sink (DDR-002) 20 °C and 85 % RH holds in rooms up to about 28 °C, with a 0.9 K dew point margin in the reference room
R3	Stability at a set point	±0.3 °C and ±2 % RH over a 30 min hold	Controller simulation; later log	Met on an idealized model: ±0.02 °C, ±1.0 % RH
R4	Uniformity across the six bays	±0.3 °C and ±2 % RH between bays	Airflow estimate; later traverse with two references	At risk: ±0.18 °C at the EPA points with the 120 mm fan, ±0.42 °C at 10 °C; ±0.8 % RH
R5	Reference temperature uncertainty	0.2 °C or better (expanded, k = 2) from 10 to 40 °C	Uncertainty budget; ice-point check	At risk: 0.14 °C with the typical sensor tolerance, 0.24 °C if it is 0.2 °C; no calibration certificate
R6	Reference humidity uncertainty	2 % RH or better (expanded) from 20 to 85 % RH, checked at four salt fixed points	Uncertainty budget; salt fixed-point checks (Greenspan, 1977)	At risk: 2.0 % RH, no margin
R7	Particle test range	Zero below 2 µg/m³ PM2.5; controlled decay from 300 to 5 µg/m³ in 45 min or less	Decay calculation; later run	Met: 5 min clean-down; 28 min decay
R8	Particle reference traceability	Transfer PM sensor collocated with a regulatory or research monitor for 30 days or more, and meeting EPA PM2.5 targets there, before use as the chamber reference	Collocation record	Not met by the rig alone. First candidate site chosen on 2026-10-02 (CLR-DEC-001): a Texas Commission on Environmental Quality site in Dallas-Fort Worth, still to be approached; no collocation done yet
R9	CO2 check	Zero check below 50 ppm with soda lime; span 400 to 2000 ppm against a reference of ±(30 ppm + 3 %)	Calculation and datasheet (Sensirion SCD30); later run	Met: 8 ppm floor; about 1,100 ppm added per liter of breath
R10	Capacity	Six sensor heads per run, each up to 90 x 70 x 50 mm, with power and I2C, UART or USB leads through a sealed gland. Large item case (CLR-DEC-001, 2026-10-02): a head up to 192	Model check	Met: six bays, 244 mm clear above the tray; large heads up to 234 mm tall fit in every two-bay and four-bay position

ID	Requirement	Target	Verification (TRL 3 or later)	Status (CLR-CAL-001)
		x 70 mm in plan takes two bays side by side, and a head up to 192 x 152 mm a 2 x 2 block of four bays, within the 244 mm clear height, with fewer heads per run		(model checks, CLR-CAL-001 [A4])
R11	Throughput	Four-point temperature and humidity sweep plus one particle decay run in 8 h or less, unattended	Timing estimate; later log	Met: about 5.5 h
R12	Cost	Within the \$412 value-engineering target for parts (<code>budget_usd</code> , a hypothetical control target; raised from \$300 to \$400 on 2026-09-25 and to \$412 on 2026-09-26 under CLR-DDR-002)	Priced BOM (<code>bom/bom.csv</code>)	Over the value-engineering target by USD 36.50: USD 448.50 estimated after the parts added to make the design buildable (CLR-DDR-003) and the front badge, door border and pull handle adopted on 2026-10-02
R13	Size and mass	Fits a 600 x 500 mm bench area, 400 mm high or less; 14 kg or less (relaxed from 12 kg under CLR-DDR-002)	Model; mass estimate	Met: 600 x 500 x 383 mm; about 13.9 kg (0.08 kg margin)
R14	Records	Every run writes CSV (time, set points, references, each sensor) and a per-sensor report based on the US EPA 2021 performance targets and testing protocol for PM2.5 air sensors, with fitted slope, offset and humidity term, EPA metrics, the reference check dates and the uncertainty chain from the salt and ice fixed points and the collocation record (CLR-DEC-001, 2026-10-02)	Software review	Not verifiable at TRL 3: report template defined (CLR-PRC-001, Table 4); software not written
R15	Electrical safety	No mains wiring in the rig; certified external 12 V supply; fused 12 V bus; peak draw 100 W or less	Design review; power budget	Met: about 90 W, 7.5 A peak
R16	Thermal safety	Independent hardware cut-off opens the heater circuit at 50 °C air temperature or 70 °C Peltier hot side	Design review; heat balance	Met: outer sink 50 °C at most in normal use
R17	Aerosol and gas safety	No toxic gases; test smoke vented through the HEPA loop before the door opens; smoke source outside the chamber	Design review	Met by design
R18	Garage-buildable	Built with a saw or laser cutter, a drill and a soldering iron; all active parts off the shelf	Design review	Met by design

■ Requirements not met or at risk

- **R8**, particle traceability. The chamber cannot make a mass reference; it transfers one from a collocation site. The first candidate to approach, chosen on 2026-10-02, is a Texas Commission on Environmental Quality site in

Dallas-Fort Worth, with South Coast AQMD's AQ-SPEC program as the fallback; R8 stays not met until the transfer sensor has been collocated there.

- **R4, R5 and R6** are at risk (uniformity at cold points, sensor tolerance and humidity budget without margin). A certified temperature probe for R5 is decided if the cost target allows; the estimated cost is already above the \$412 value-engineering target, so it does not yet.
- R2 and R13 moved to met in v0.4 (larger inner sink; mass limit relaxed to 14 kg). R12 moved to within the value-engineering target in v0.5: the target was set by Amish on 2026-09-26 at \$412, which covered the priced BOM then.
- **R12**, cost, moved in v0.6 to over the value-engineering target by \$27: the front frame, bulkhead fittings, cable glands, drain and printed holders that make the design buildable (CLR-DDR-003) brought the estimated cost to \$439. In v0.9 the front badge, door border and pull handle adopted on 2026-10-02 add \$9.50. Value-engineering target: USD 412. Estimated cost of the constructable design: USD 448.50 (USD 36.50 over the target). The target is a hypothetical control target, not a limit; savings worth trying are in the value engineering section of CLR-DEC-001. R13 stays met at about 13.9 kg with a 9 mm base, with a 0.08 kg margin.
- **R10** moved in v0.9 from met for six heads to met including the large item case, checked in the model.

■ Assumptions

- Chamber conductance about 0.82 W/K with the full jacket and door panel fitted, and 6.7 W of internal gains (CLR-CAL-001).
- Settling is limited by the 3 L/min conditioning flow, moisture held on the walls and a 15 min equilibration allowance for the heads.
- Test aerosol is incense smoke (CLR-DDR-001 A4); results for other aerosols (dust, sea salt) will differ and must be stated in each report.